

```
1  from queue import PriorityQueue
2
3  g = {'A': [('B', 1), ('C', 3)], 'B': [('D', 3)
    ], 'C': [('G', 5)], 'D': [('G', 2)],
    'G': []}
4  h = {'A': 5, 'B': 3, 'C': 4, 'D': 2, 'G': 0}
5
6  def astar(s, goal):
7      q = PriorityQueue()
8      q.put((0, s, [s]))
9
10     while not q.empty():
11         f, n, path = q.get()
12         if n == goal:
13             print("Path:", " -> ".join
                (path))
14             return
15         for x, c in g[n]:
16             q.put((c + h[x], x, path +
                [x]))
17
18  astar('A', 'G')
```

Path: A -> B -> D -> G

=== Code Execution Successful ===